Section 2

INDUCTION MOTOR (IM)

Simulation Test 1

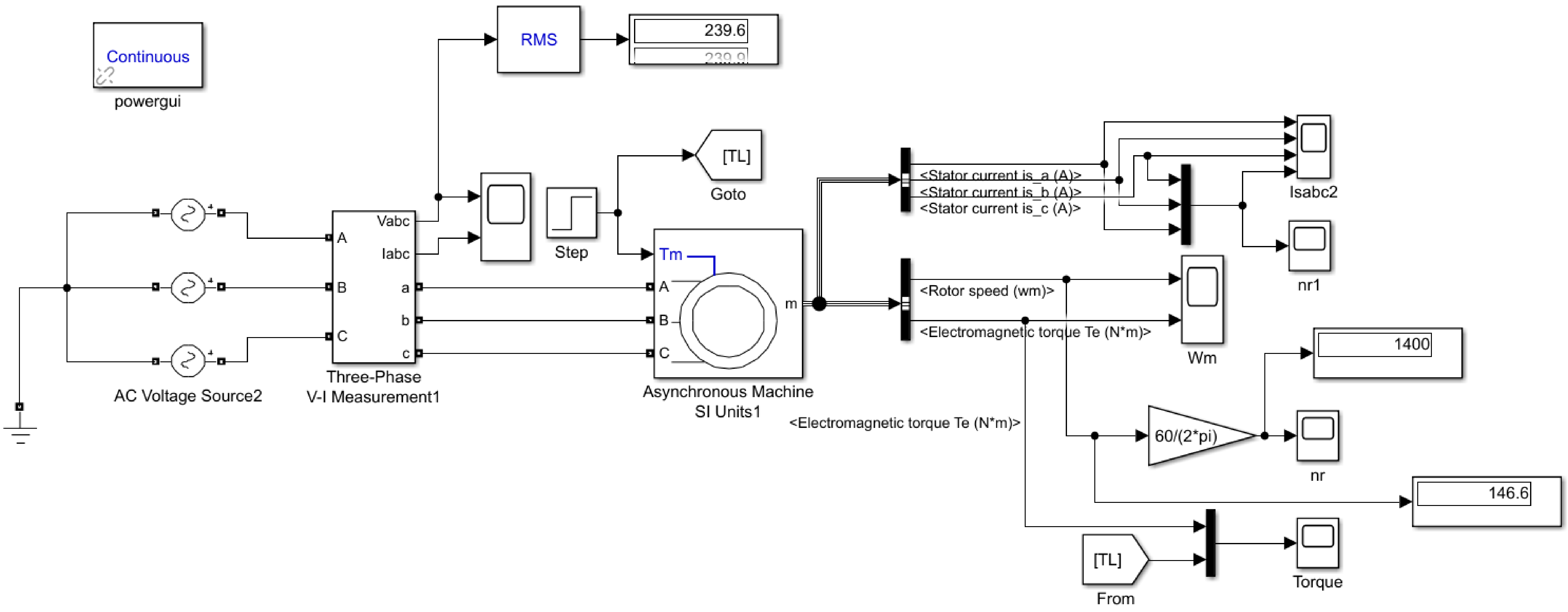
MATLAB
SIMULINK

Using Matlab/Simulink to create:

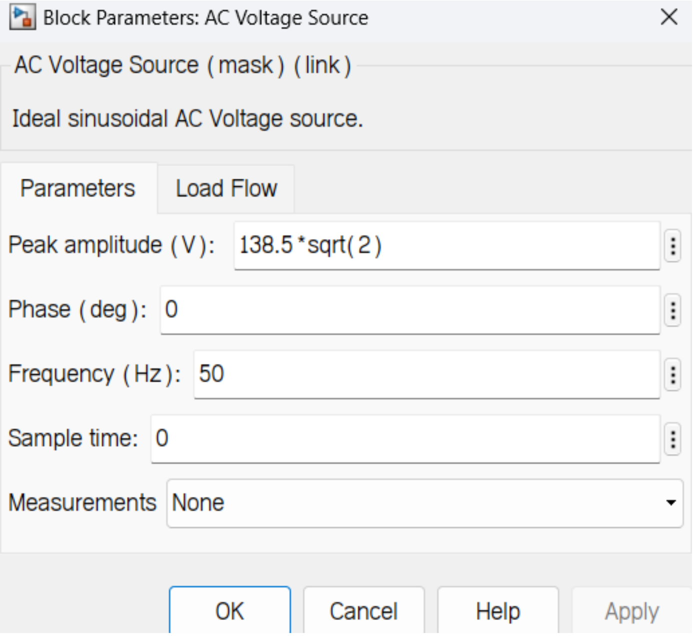
Open loop of an Induction Motor System

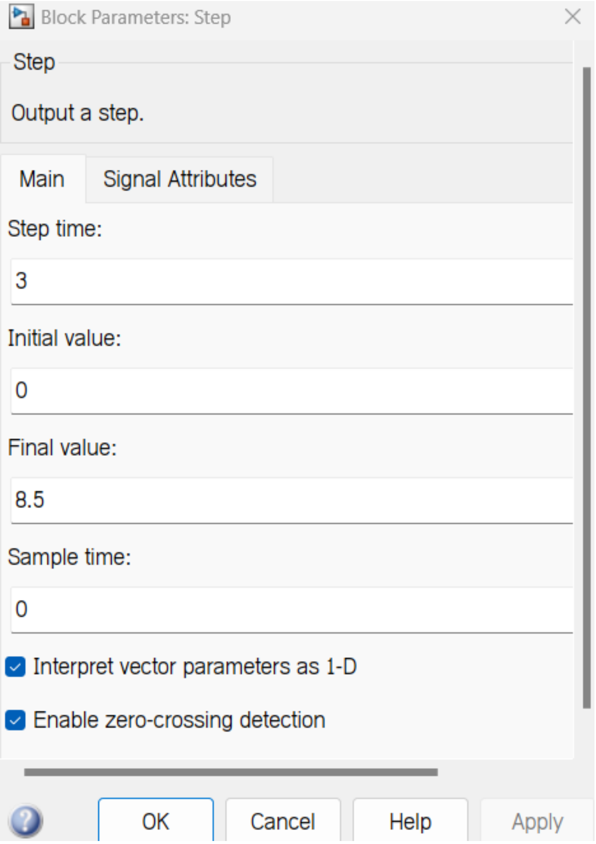
Question

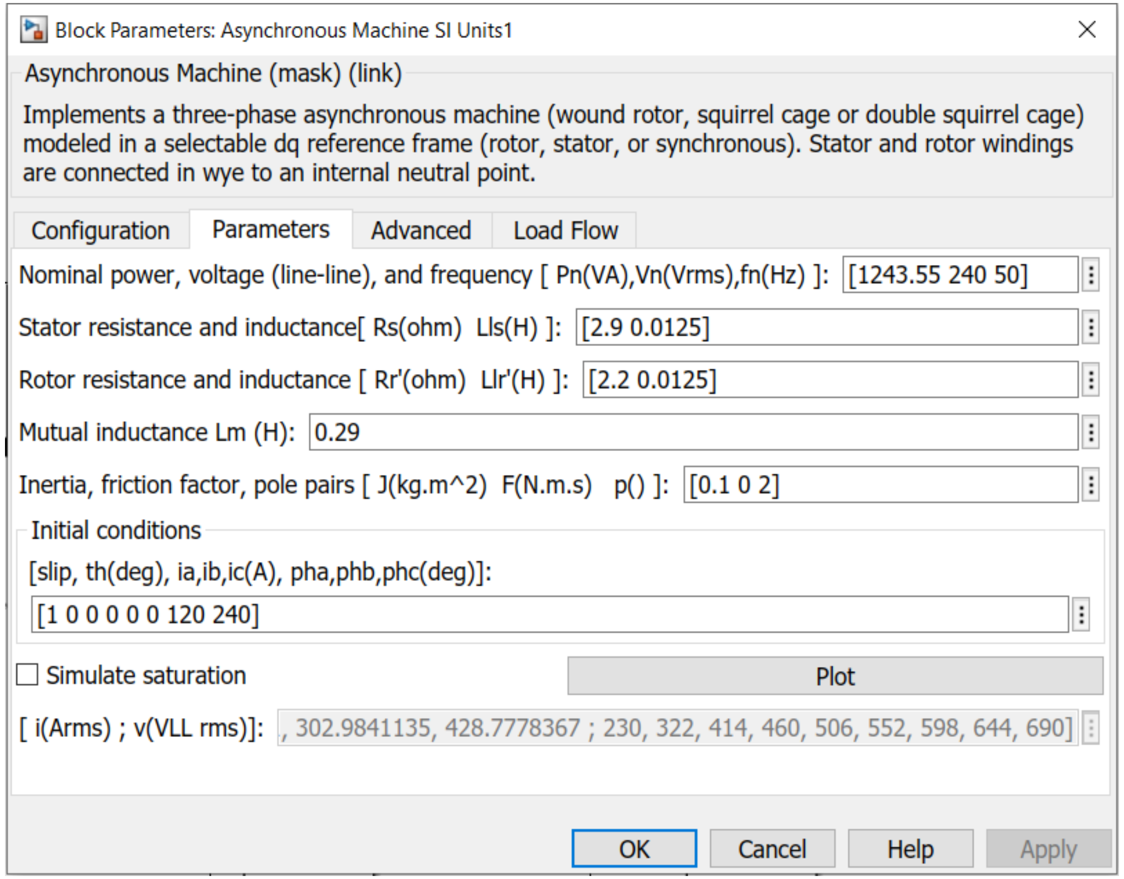
1. Plot speed, torque, voltage and current of the motor.
2. Determine slip of the motor after apply the load torque?

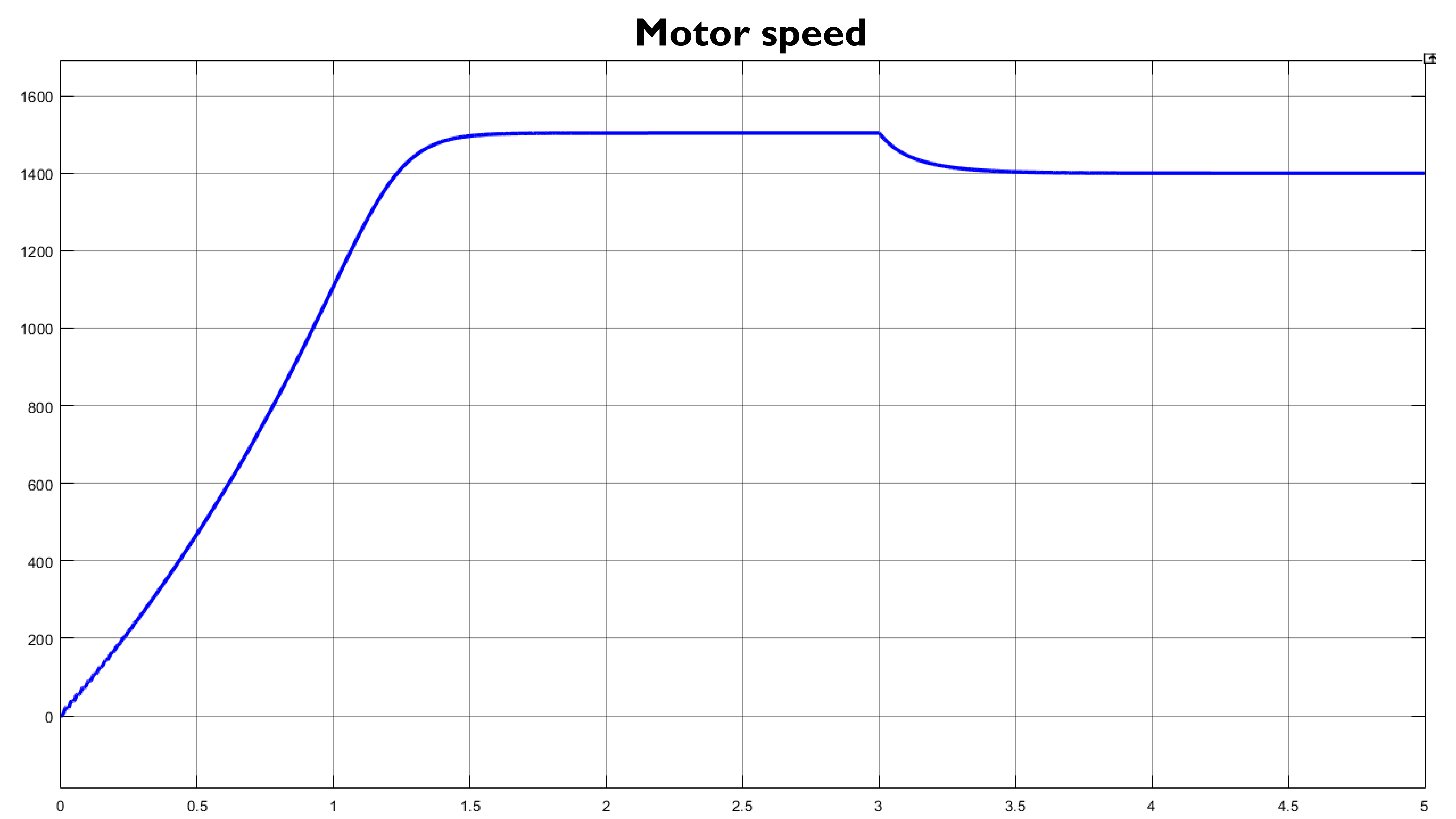


Continuous
powergui
RMS
239.6
239.9
[TL]
Goto
Step
Vabc
Iabc
A
B
C
a
b
c
Three-Phase
V-I Measurement1
AC Voltage Source2
Tm
m
Asynchronous Machine
SI Units1
<Stator current is_a (A)>
<Stator current is_b (A)>
<Stator current is_c (A)>
Isabc2
nr1
<Rotor speed (wm)>
<Electromagnetic torque Te (N*m)>
Wm
1400
<Electromagnetic torque Te (N*m)>
60/(2*pi)
nr
146.6
[TL]
From
Torque

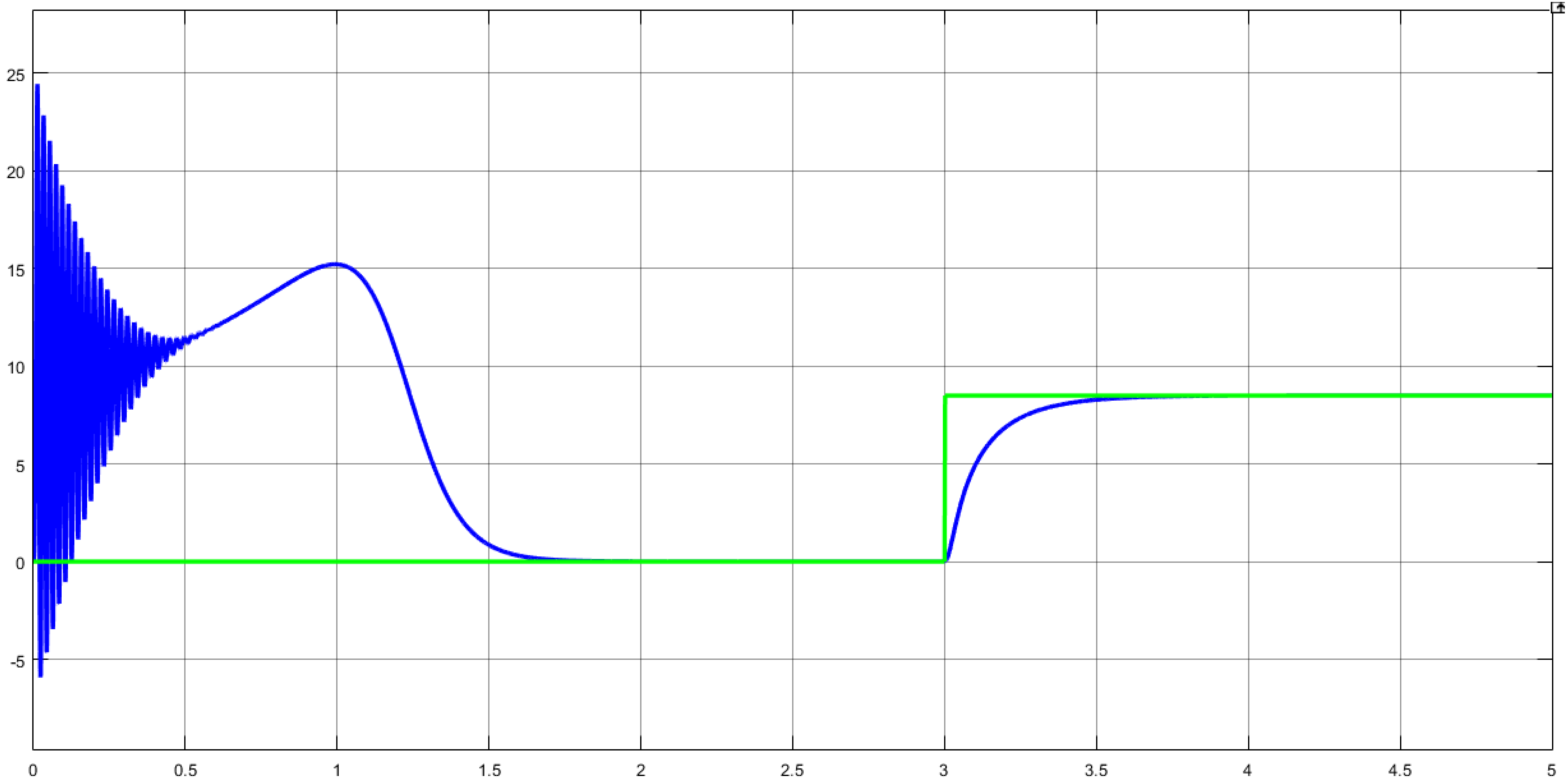

Block Parameters: AC Voltage Source
AC Voltage Source (mask) (link)
Ideal sinusoidal AC Voltage source.
Parameters
Load Flow
Peak amplitude (V): 138.5*sqrt(2)
Phase (deg): 0
Frequency (Hz): 50
Sample time: 0
Measurements None
OK
Cancel
Help
Apply


Block Parameters: Step
Step
Output a step.
Main
Signal Attributes
Step time:
3
Initial value:
0
Final value:
8.5
Sample time:
0
Interpret vector parameters as 1-D
Enable zero-crossing detection
OK
Cancel
Help
Apply


Block Parameters: Asynchronous Machine SI Units1
Asynchronous Machine (mask) (link)
Implements a three-phase asynchronous machine (wound rotor, squirrel cage or double squirrel cage) modeled in a selectable dq reference frame (rotor, stator, or synchronous). Stator and rotor windings are connected in wye to an internal neutral point.
Configuration
Parameters
Advanced
Load Flow
Nominal power, voltage (line-line), and frequency [Pn(VA),Vn(Vrms),fn(Hz)]: [1243.55 240 50]
Stator resistance and inductance[Rs(ohm) Lls(H)]: [2.9 0.0125]
Rotor resistance and inductance [Rr'(ohm) Llr'(H)]: [2.2 0.0125]
Mutual inductance Lm (H): 0.29
Inertia, friction factor, pole pairs [J(kg.m^2) F(N.m.s) p()]: [0.1 0 2]
Initial conditions
[slip, th(deg), ia,ib,ic(A), pha,phb,phc(deg)]:
[1 0 0 0 0 0 120 240]
Simulate saturation
Plot
[i(Arms) ; v(VLL rms)]: , 302.9841135, 428.7778367 ; 230, 322, 414, 460, 506, 552, 598, 644, 690]
OK
Cancel
Help
Apply


Motor speed
1600
1400
1200
1000
800
600
400
200
0
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5

Motor Torque



Voltage:

Current: